OPERATIONS MANUAL & SYSTEM ARCHITECTURE

STANDARD OPERATING PROCEDURES & SYSTEM MANUAL

Retail Operating Standards, Digital Transaction Governance, 24 Solar Terms Botanical Formulations, and Cloud E-Commerce Cafe Architecture.

DOCUMENT IDENTIFICATION & METADATA	
Document Identification Code	`DOC-CKR-OPS-2026-V2-EN`
Classification & Distribution	Official Operating Standard (Internal Staff & Academic/Business Review)
Release Version & Effective Date	Version 2.1 (Comprehensive Editorial Revision) — October 2026
Publishing Authority	Board of Founders, CHANGKEER ATELIER (5 Co-Founders)

CHANGKEER BOARD OF FOUNDERS (THE 5 CO-FOUNDERS)

No	Founder Name	Role	Executive Operational Focus
01	**ALMAIDAH CHAERUDIN PUTRI**	Co-founder	Championing honest daily hydration without synthetic shortcuts.
02	**ANNISHA HUMAIRA**	Co-founder	Formulating whole-flower blends aligned with seasonal body rhythms.
03	**JOHANA ABIGAEL RUMBIAK**	Co-founder	Shaping mindful, plastic-free rituals that fit active lifestyles.
04	**KEYLA PUTRI ADISTY**	Co-founder	Fostering radical transparency and honest ingredient education.
05	**NURSOLEHA ROMADHONI**	Co-founder	Partnering directly with local growers across Puncak, Garut & Wonosobo.

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Executive Summary

Document Structure & Table of Contents

This document serves as an authoritative operating reference for atelier staff, baristas, retail cashiers, technology operators, and academic/business evaluators. All content is organized systematically and supported by rigorous technical and operational evidence.

Chapter	Chapter Title & Scope of Coverage	Core Operational Focus
CHAPTER 01	**Concept, Business Identity & Vision**	Brand profile, 24 Solar Terms (Jieqi) philosophy, tisane rationale, and the 5 Co-Founders structure.
CHAPTER 02	**System Architecture & Technology Ecosystem**	End-to-end system workflow, modern tech stack, dual-tier fault tolerance, and offline cache resilience.
CHAPTER 03	**Complete Catalog of 24 Seasonal Blends**	Comprehensive formulation tables across 4 seasons, water temperatures, official pricing, and customization.
CHAPTER 04	**Transaction Governance & Dynamic QRIS**	Linear ordering workflow, dynamic QRIS security, server-side anti-tampering, and idempotency locks.
CHAPTER 05	**Barista Standard Operating Procedures (SOP)**	KDS touch-monitor ticket handling, precision brewing temps and timing, calibrated sweetness, and bar hygiene.
CHAPTER 06	**Cashier Standard Operating Procedures (SOP)**	Counter POS order flow, payment troubleshooting matrix, refund policies, and daily cash reconciliation.
CHAPTER 07	**Store Administration & Layman Glossary**	Rapid inventory stock toggling (/admin) and accessible glossary for non-technical evaluators.

Chapter 01 • Business Concept & Identity

Understanding CHANGKEER & The 24 Seasons

1.1 Enterprise Profile & Core Values

CHANGKEER ATELIER is a contemporary artisan herbal tea atelier and botanical laboratory that harmonizes the indigenous medicinal flora of the Indonesian archipelago with the ancient wisdom of the 24 Solar Terms (*24 Jieqi*). Unlike fast-paced commercial coffee chains high in stimulants or conventional folk medicine stalls lacking standardization, CHANGKEER provides a serene sanctuary for mindful hydration, calibrated extraction, and holistic well-being.

Botanically, all CHANGKEER infusions are classified as **Tisanes**—pure extractions of whole culinary herbs, rhizomes, and botanical flower petals—entirely free of black tea leaves, synthetic flavoring essences, and refined chemicals. This ensures that every formulation is gentle on sensitive digestion and completely non-stimulant, making it suitable for any hour of the day without disrupting sleep quality.

1.2 The Philosophy of 24 Solar Terms (Jieqi)

Human biology exhibits clear circadian and seasonal responses to climatic transitions. During scorching heatwaves, internal organs benefit from botanical agents that disperse metabolic heat. Conversely, during torrential monsoons and cold drafts, volatile essential oils from rhizomes are needed to stimulate peripheral blood circulation and prevent physical fatigue.

CHANGKEER organizes its 24 signature formulations across four primary seasonal pillars:

- **Spring Season (春季):** Light floral compositions that awaken vital organ function and gently cleanse metabolic residues after seasonal dormancy.
- **Summer Season (夏季):** Mountain peppermint, butterfly pea, and native citrus designed to lower core temperature and provide thirst-quenching hydration.
- **Autumn Season (秋季):** Whole chamomile blossoms, tropical cinnamon bark, and sun-dried pears to soothe psychological tension and promote deep sleep.
- **Winter Season (冬季):** Aged red ginger, cloves, and wild raw honey to deliver penetrating internal warmth and bolster immune resilience.

1.3 The Rationale for a Self-Service Digital Platform

Brewing premium botanical tisanes requires absolute precision regarding water temperature and infusion timing. If baristas are repeatedly interrupted by taking manual orders on paper chits or making change with cash, beverage quality suffers immediately. CHANGKEER's self-service web ordering platform empowers guests to customize their brews directly from their table via smartphone, enabling baristas to dedicate 100% of their focus to artisanal craftsmanship.

Chapter 02 • System Architecture

System Architecture & Technology Ecosystem

2.1 Integrated Architectural Principles

The CHANGKEER software platform is engineered with modern web standards, prioritizing instant loading speeds, banking-grade transactional integrity, and effortless bar-station operation. Guests access the entire menu directly in their mobile browser without downloading native apps.

2.2 Selected Technology Stack

Architectural Tier	Chosen Technology	Engineering Justification & Operational Benefit
Primary Web Application	Next.js 16 (React 19, App Router)	Instant mobile page renders with zero bundle bloat; fast server-side generation for static menu catalogs.
Programming Language	TypeScript 5 (Strict Mode)	Eliminates runtime type errors during development, safeguarding financial math, voucher logic, and recipe data.
User Interface & Styling	Tailwind CSS	Ergonomic touch targets, clean typographic hierarchy, zero extraneous decoration, and full responsive behavior across devices.
Database & Data Layer	PostgreSQL (via Supabase)	Relational consistency with full ACID compliance and Row-Level Security (RLS) to protect transactional records and customer privacy.
Data Validation Layer	Zod Schema Validation	Guarantees that every incoming HTTP order payload matches exact schema constraints before server execution.

2.3 Dual-Tier Fault Tolerance & Offline Resilience

To protect retail store operations against unexpected internet interruptions, the architecture features a dual-tier storage strategy. Active table orders are mirrored in high-resilience local cache memory at the bar terminal, guaranteeing that the barista ticket queue and receipt printing function seamlessly even during severe ISP connectivity lag.

Chapter 03 • Product Catalog

Complete Catalog of 24 Seasonal Blends

Below is the authoritative documentation of ingredient formulations, water extraction standards, and official retail pricing for all 24 CHANGKEER ATELIER botanical tisanes:

Spring Season (春季) — Solar Terms 01 to 06

Characterized by fresh, light floral notes and young botanical shoots to awaken vitality and support natural metabolic renewal.

No	Solar Term	Blend Formulation	Primary Ingredients	Brew Standard	Price	Flavor Profile & Notes
01	 **Start of Spring** 立春	**First Bloom**	jasmine flowers, dried pear, goji	82°C • 3.5 Min	Hot 24K Iced 26K	Light, floral and a little fruity.
02	 **Rain Water** 雨水	**Rain Garden**	rose buds, ginger, red dates	98°C • 6.0 Min	Hot 20K Iced 22K	Rose up front, soft ginger warmth.
03	 **Awakening of Insects** 惊蛰	**Green Wake**	mint, lemongrass, pandan	88°C • 4.0 Min	Hot 15K Iced 17K	Fresh and grassy, with a pandan finish.
04	 **Spring Equinox** 春分	**Even Day**	chrysanthemum, rose buds, goji	92°C • 5.0 Min	Hot 22K Iced 24K	Half floral, half fruity.
05	 **Clear and Bright** 清明	**Clear Sky**	butterfly pea, lemongrass, lemon	88°C • 4.0 Min	Hot 18K Iced 20K	Blue in the cup, purple once the lemon goes in.
06	 **Grain Rain** 谷雨	**Barley Field**	roasted barley, osmanthus	88°C • 4.0 Min	Hot 20K Iced 22K	Toasty and nutty, with a honeyed flower note.

Summer Season (夏季) — Solar Terms 07 to 12

Featuring cooling mountain mint, native citrus, and soothing flowers designed for active hydration and biological heat dispersion.

No	Solar Term	Blend Formulation	Primary Ingredients	Brew Standard	Price	Flavor Profile & Notes
07	 **Start of Summer** 立夏	**Hibiscus Sun**	hibiscus, orange peel, mint	92°C • 5.0 Min	Hot 18K Iced 20K	Tart, ruby-red and very refreshing.
08	 **Grain Buds** 小满	**Pandan Lime**	pandan, lemongrass, lime leaf	88°C • 4.0 Min	Hot 15K Iced 17K	Green and citrusy.
09	 **Grain in Ear** 芒种	**Mango Jasmine**	dried mango, jasmine flowers	82°C • 3.5 Min	Hot 26K Iced 28K	Ripe fruit with a jasmine lift.
10	 **Summer Solstice** 夏至	**Longest Day**	chrysanthemum, lemon, mint	88°C • 4.0 Min	Hot 18K Iced 20K	Bright, clean and cooling.
11	 **Minor Heat** 小暑	**Cool Leaf**	mint, lime leaf, lemongrass	88°C • 4.0 Min	Hot 15K Iced 17K	Minty and sharp.
12	 **Major Heat** 大暑	**Chrysanthemum Lemon**	chrysanthemum, lemon, goji	88°C • 4.0 Min	Hot 20K Iced 22K	Our hot-season classic: mellow flowers, fresh citrus.

Autumn Season (秋季) — Solar Terms 13 to 18

Rich in sweet cinnamon bark, whole chamomile blossoms, and roasted grains to ground the nervous system and promote restorative evening calm.

No	Solar Term	Blend Formulation	Primary Ingredients	Brew Standard	Price	Flavor Profile & Notes
13	 **Start of Autumn** 立秋	**Osmanthus Pear**	osmanthus, dried pear, goji	88°C • 4.0 Min	Hot 26K Iced 28K	Apricot-scented flowers with soft pear.
14	 **End of Heat** 处暑	**Late Rose**	rose buds, hibiscus, dried apple	92°C • 5.0 Min	Hot 22K Iced 24K	Rosy and tart, like fruit jam.
15	 **White Dew** 白露	**White Dew**	white chrysanthemum, dried pear	88°C • 4.0 Min	Hot 22K Iced 24K	Delicate and smooth. Try it with honey.
16	 **Autumn Equinox** 秋分	**Golden Hour**	osmanthus, roasted barley, orange peel	88°C • 4.0 Min	Hot 24K Iced 26K	Toasty and golden.
17	 **Cold Dew** 寒露	**First Rain**	ginger, goji, red dates	98°C • 6.0 Min	Hot 20K Iced 22K	Warming ginger and naturally sweet dates, for the first wet evenings.
18	**Frost** 霜降	**Ginger Rose**	ginger, rose buds, cinnamon	98°C • 6.0 Min	Hot 20K Iced 22K	Spiced and floral.

Winter Season (冬季) — Solar Terms 19 to 24

Anchored by mature red ginger, aromatic archipelago cloves, and wild forest honey to provide deep warmth and sustain immune resilience.

No	Solar Term	Blend Formulation	Primary Ingredients	Brew Standard	Price	Flavor Profile & Notes
19	 **Start of Winter** 立冬	**Spice Lantern**	cinnamon, clove, dried apple	98°C • 6.0 Min	Hot 22K Iced 24K	Baked-apple warmth with a little clove.
20	 **Minor Snow** 小雪	**Longan Nest**	longan, red dates, goji	88°C • 4.0 Min	Hot 28K Iced 30K	Caramel-sweet fruit, no sugar added.
21	 **Major Snow** 大雪	**Ginger Goji**	ginger, goji, lemon	98°C • 6.0 Min	Hot 18K Iced 20K	The rainy-season favourite.
22	 **Winter Solstice** 冬至	**Solstice Cup**	red dates, ginger, osmanthus	98°C • 6.0 Min	Hot 24K Iced 26K	Deep, warm and gently floral.
23	 **Minor Cold** 小寒	**Red Date Warmth**	red dates, longan, ginger	98°C • 6.0 Min	Hot 26K Iced 28K	Rich and mellow.
24	 **Major Cold** 大寒	**Cinnamon Ember**	cinnamon, ginger, orange peel	98°C • 6.0 Min	Hot 18K Iced 20K	Spicy-sweet, the last blend before spring.

3.2 Standardized Customization Options

Guests may calibrate each brew to their personal preferences according to the following official parameters:

Customization Axis	Available Options	Operational Notes & Surcharge
Serving Temperature	Hot Ceramic Tea Pot / Iced Shaken Tea	Traditional ceramic hot infusion (standard) or craft iced shaken tea with crystal ice cubes (+IDR 2,000).
Natural Sweetener Level	0%, 30%, 50%, 75%, 100%	Accurately calibrated wild honey or pure sorghum nectar dispensed via jigger. Completely free of artificial sugars.
Dried Flower Toppings	Rose Buds, Chrysanthemum, Butterfly Pea, Jasmine, Osmanthus, Hibiscus (+IDR 3,000 – 4,000)	Whole food-grade dried floral petals to elevate volatile aromas, visual infusion colors, and relaxation.
Sun-Dried Fruit Toppings	Goji Berries, Red Dates / Jujube, Longan, Pear, Apple, Mango, Lemon, Orange Peel (+IDR 2,000 – 4,000)	Unsweetened dehydrated fruit slices releasing wholesome natural fructose and vibrant fruit depth.

Chapter 04 • Payment Security

Transaction Governance & Dynamic QRIS

4.1 End-to-End Digital Ordering Workflow

The transactional pipeline is strictly linear, spanning from guest table selection to beverage handover at the preparation counter:

STEP 1
Guest Selects Menu
Select blend & customize in mobile browser

→

STEP 2
System Locks Total
Server generates dynamic, exact-amount QRIS

→

STEP 3
Payment Verified
Banking gateway confirms settlement

→

STEP 4
Ticket Sent to KDS
Recipe details appear on barista monitor

→

STEP 5
Barista Crafts Brew
Precision extraction according to SOP

→

STEP 6
Order Handover
Tea served & ticket marked complete

4.2 Comparative Evaluation: Static Paper QRIS vs CHANGKEER Dynamic QRIS

Evaluation Metric	Static QRIS (Conventional Printed Sticker)	CHANGKEER Dynamic QRIS
Amount Input	Manually entered by guest; high risk of typing errors (omitted or added zeros).	Cryptographically locked by server; exact bill amount rendered automatically.
Verification Speed	Cashier must physically inspect guest smartphone screen, causing checkout queues.	Automated webhook confirmation; order updates across screens in under 2 seconds.
Fraud & Fake Slip Risk	High; vulnerable to manipulated bank receipt screenshots or payment app simulators.	Zero; server accepts fulfillment only upon direct webhook verification from payment gateway.

4.3 Server-Side Anti-Tampering Security

Many retail web applications suffer fraud from malicious browser parameter manipulation (e.g., using developer inspect tools). In CHANGKEER's architecture, unit prices **are never trusted from the client device**. Upon order submission, the server re-validates each item ID directly against the central master catalog. If any price discrepancy or tampering is detected, the transaction is immediately aborted by the security gateway.

4.4 Double-Debit Prevention (Idempotency Key)

When a guest encounters poor mobile reception and presses the pay button multiple times in rapid succession, the platform engages an *idempotency lock*. Each transaction is tagged with a unique cryptographic idempotency token ensuring that the underlying banking gateway will execute settlement exactly once.

Chapter 05 • Barista Standards

Barista Standard Operating Procedures (SOP)

SOP-BAR-01: Kitchen Display System (KDS) Ticket Management

Purpose: Ensure all beverage orders are prepared strictly in FIFO queue order without confusion or delay.

Operational Steps:

1. Monitor the high-contrast KDS touch-screen stationed directly above the brewing counter.
2. Identify new tickets marked with **'Order Received' (Amber)** and review recipe specifications: table number, blend name, water temperature, and sweetness level.
3. Press the **'Start Brew'** button on the touch interface. The ticket color switches to **'Brewing Now' (Blue)** and the digital countdown timer begins.
4. Upon timer completion, pour the finished infusion into the designated pre-warmed ceramic pot or iced shaker, then tap **'Ready to Serve'** to switch status to **'Ready' (Green)**.

> **Mandatory Rule:** Baristas must not bypass queue order unless explicitly instructed by the Shift Supervisor during peak rush management.

5.2 Water Temperature & Extraction Timing Standards

Different botanical tissue structures require specific water thermal envelopes to extract beneficial volatile oils without releasing bitter polyphenols:

Botanical Classification	Ideal Water Temp	Extraction Time	Quality Assurance Guidelines
Fresh Floral & Green Tea	**80°C – 85°C**	3.5 – 4.0 Min	Use digital gooseneck variable-temp kettles; avoid boiling water to prevent thermal scorching of jasmine petals.
Dried Blossoms (Chamomile, Roselle)	**90°C – 94°C**	5.0 Min Full	Ensure petals are fully submerged to allow complete release of natural pigments and calming aromatics.
Rhizomes & Roots (Red Ginger, Turmeric)	**98°C – 100°C**	6.0 Min Full	Full rolling boil required to break dense cellular walls and release therapeutic gingerols and essential oils.
Aromatic Bark & Whole Seeds	**95°C – 98°C**	5.5 Min Full	Keep ceramic infusion lids firmly closed throughout extraction to prevent fragrant aromatic vapors from escaping.

SOP-BAR-02: Calibrated Natural Sweetener Dispensing

Purpose: Maintain precise flavor consistency across all sweetness levels using calibrated stainless jiggers.

Standard Jigger Dispensing Matrix:

Sweetness Level	Jigger Measure (ml)	Flavor Balance & Beverage Character
0% (Unsweetened)	0 ml	Pure herbal formulation; crisp, authentic botanical profiles with zero added sugar.
30% (Light Sweetness)	7.5 ml	Delicate honey finish on the palate; floral notes remain distinctly dominant.
50% (Balanced / Standard)	12.5 ml	Atelier benchmark; harmonious equilibrium between herbal notes and natural nectar sweetness.
75% (Enhanced Sweetness)	18.0 ml	Sweetness is prominent; recommended for tart hibiscus blends and iced shaken varieties.
100% (Full Sweetness)	25.0 ml	Rich and full-bodied; entirely throat-friendly thanks to 100% natural raw nectar sources.

> **Mandatory Rule:** Baristas must never pour sweeteners by eye. Always employ certified stainless steel jiggers.

Chapter 06 • Cashier Standards

Cashier Standard Operating Procedures & Reconciliation

6.1 Counter Cashier Responsibilities

While the majority of guests complete transactions autonomously via mobile QRIS at tables, counter cashiers play a critical role in assisting walk-in guests requesting cash settlement, operating counter POS terminals, troubleshooting bank gateway delays, and executing end-of-day store closing balances.

6.2 Payment Troubleshooting Matrix for Cashiers

Operational Scenario	Underlying Root Cause	Standard Action Protocol
QRIS Code Fails to Scan	Guest screen brightness too low or camera lens smudged.	Politely guide guest to increase screen brightness to 100%. If scanning continues to fail, generate an instant order chit at the counter POS.
Funds Debited, Status Stalled	National interbank clearing latency during peak processing hours.	Verify successful debit notification on guest's banking app. Cross-reference transaction ID, then press 'Manual Settle' on the cashier terminal.
Accidental Menu Item Selection	Guest confirmed cart hurriedly on personal device.	Provided the barista has not yet tapped 'Start Brew' on the KDS, the cashier may cancel the item in the terminal and issue the correct replacement.

SOP-CAS-01: End-of-Day Financial Reconciliation (Shift Closing)

Purpose: Ensure 100% balance matching between physical drawer cash, digital QRIS settlements, and system sales totals.

Closing Procedure Steps:

1. At 21:45 WIB, lock order intake and print the daily executive sales summary from the `/admin` portal.
2. Remove all physical currency from the cash drawer and count bank notes and coins systematically in the presence of the Shift Supervisor.
3. Segregate the initial float fund (IDR 500,000) and place it securely into the designated float envelope.
4. Compare remaining net physical cash against the 'Total Cash Revenue' figure reported by the POS system.
5. Review the merchant payment gateway settlement batch report and verify against 'Total Digital Revenue'.
6. Co-sign the Daily Reconciliation Log alongside the Shift Supervisor and deposit the reconciled cash into the store safe.

> **Variance Tolerance:** Cash variance policy is strict IDR 0 tolerance. Any discrepancy exceeding IDR 2,000 mandates an immediate item-by-item receipt audit.

Chapter 07 • Store Administration & Glossary

Store Administration & Layman Practical Glossary

7.1 Rapid Botanical Stock Management (/admin)

Botanical harvest yields from partner tea gardens in Wonosobo, Garut, and Puncak fluctuate with mountain weather. Should a botanical ingredient deplete unexpectedly during trading hours, store supervisors can instantly toggle item availability:

1. Access the administrative console at `/admin/inventory` using authorized supervisor credentials.
2. Locate the target blend affected by the ingredient stock-out.
3. Toggle the availability switch from **'Available'** to **'Out of Stock'**.
4. Within 1 second, the item is grayed out across all table menus, preventing guest disappointment and ordering conflicts.

7.2 Practical Glossary for Non-Technical Evaluators

The table below provides straightforward, non-technical explanations for technological and botanical terminology utilized throughout CHANGKEER ATELIER:

Technical / Herbal Term	Plain-Language Explanation	Real-World Atelier Application
Dynamic QRIS	A smart banking payment barcode that comes pre-loaded with the exact bill amount, eliminating manual typing.	A guest scans the table barcode and their banking app immediately shows the exact IDR 32,500 bill without typing errors.
Kitchen Display System (KDS)	A digital touchscreen order queue positioned at the brewing bar, replacing paper order tickets prone to liquid damage.	The bar monitor automatically alerts the barista that Table 4 has ordered a Jasmine infusion at 50% sweetness.
Idempotency Lock	A banking protection mechanism ensuring that a customer's account is never charged twice if the pay button is tapped repeatedly.	A guest with sluggish internet taps 'Pay' three times; the bank gateway debits their account exactly once.
Server-Side Anti-Tampering	A security system preventing malicious actors from altering drink prices through browser code modification.	If an attacker alters a drink price from IDR 26,000 to IDR 1,000 in mobile code, the central server rejects the order instantly.
Botanical Tisane	Pure hot or cold infusions made exclusively from herbs, flowers, and spices without ordinary black tea leaves or caffeine.	Refreshing red roselle or soothing honey chamomile that can be enjoyed late at night without heart palpitations.
Solar Terms (24 Jieqi)	The traditional division of the solar year into 24 astronomical points, matching herbal properties to bodily weather needs.	Serving a cooling mint infusion during the peak midday heat of Summer Solstice (夏至).

OFFICIAL OPERATIONAL DOCUMENT — CHANGKEER ATELIER

Published by the Board of Founders: Almaidah Chaerudin Putri, Annisha Humaira, Johana Abigael Rumbiak, Keyla Putri Adisty, Nursoleha Romadhoni.

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